DEPARTMENT OF MECHANICAL ENGINEERING

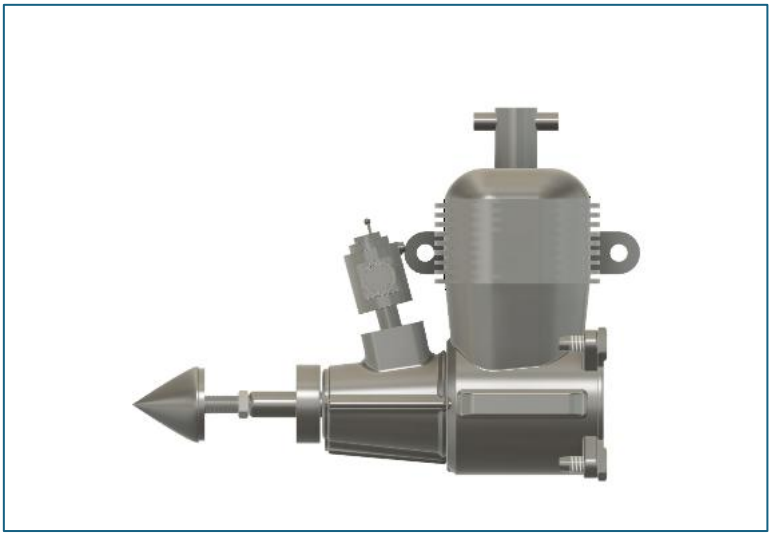
Small-scale two-stroke, compression ignition engine – Manufacturing Report

Mechanical Design A (04 23808)

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Appendix A: University of Birmingham Assessment and Feedback Student Template

Section One: Reflection on Previous Feedback (Areas for Improvement Identified)

Based on feedback received from previous assessments, the following key areas for improvement were identified:

- Improve the clarity and structure of technical explanations to ensure they are concise, logical, and easily understood by the reader.
- Improve the justification of design choices and material selection, supported by engineering principles and relevant standards.
- Improve the use of figures and diagrams so that they directly support and reinforce the written discussion.

Section Two: Actions Taken in This Assignment in Response to Previous Feedback

In this assignment, the following steps have been taken to address the feedback received previously:

- The report has been restructured to improve logical flow, with clearer sectioning and progression of ideas to enhance readability.
- Design, material, and manufacturing decisions are explained more clearly using engineering reasoning, constraints, and functional requirements.
- CAD models, drawings, and figures are used more effectively, with clearer labelling and direct references within the text.

Section Three: Feedback Requested on This Assignment

Feedback on the following aspects of this assignment would be particularly valuable for further improvement:

- The adequacy of the manufacturing detail and process explanations relative to the module requirements.
- The clarity and effectiveness of the technical explanations in communicating design intent and engineering reasoning.
- The appropriateness of the level of detail across different sections of the report.

Appendix B: Report writing checklist

Aspect	Checked
Cover page: Completed as per requirements. · Margins are correct as per requirements. If not given, margin has been picked. · Font size is as per the requirements. If not sure, then large enough to read. · Use of capital letters has been checked. · Line spacing is as per the requirements. If not sure, then default is used.	☑
· All text and writing in English as required on the submission sheet.	☑
· Figures and tables are all introduced in the text before they appear in the report.	☑
· All figures or tables have a caption describing what the figure/table shows. i.e. Figure 1: Description of…Normally figures are below, tables are above. If using a), b), etc. as sub parts of a figure, then the over figure is introduced first then the sub parts. i.e. Figure 1: Description of…. a) casting, b) milling, c) grinding.	☑
· Figure and table detailed in the text has a capital letter i.e., ….as shown in Figure 1. - Have captions	☑
- Follow guidelines above for rest of appendix. - References are consistent in format, and everyone has been checked for this.	☑
· Figures and tables have a large enough font size for easy reading.	☑
· Figures and tables are large enough to understand the points we wish to raise.	☑
· Figures are annotated to illustrate the key points using text with a larger enough font size and clear arrows. · No figures or tables have text wrapped round them.	☑
· Figures and tables do not go over two pages. If this is required because they are too large, then they are separate figures or tables.	☑
· Figures and tables appear after the paragraph that is detailing them. Not mid paragraph. · Section headings are included. These have been checked for consistency between them. Font size and layout.	☑
· Section headings are not on a different page to the paragraph they appear in. · Page numbers are detailed (if required).	☑
· Spelling has been checked carefully. This includes tables as often not checked.	☑
· Grammar has been checked carefully. · Appendices have been treated in the same way as the report. - Titles detailing what they show.	☑
- Figures and tables are introduced before they appear.	☑
- Layout is clear and detailed.	☑

Executive Summary

The focus of this report is to examine design modifications, manufacturing processes and problems of production in a small-scaled, single-cylinder, two-stroke diesel engine that will be applied to unmanned aerial vehicles (UAVs). The design is based on the materials, scalability as well as environmental considerations of the target produce of 50,000 units. The engine is ended up being 0.76cc with a bore of 10.49mm and a stroke of 8.74mm. The engine is made to be able to rev up to 12,000 rpm and the fuel under consumption is diesel due to its density and the purpose under which it will be used in UAV operation where power consumption is the main concern and power is needed.

The structural parts of diesel engines including crankcase and crankshaft are made by using AISI 4340 steel alloy, which is characterized by stability and strength, and thus could be used under high pressure. The factors of manufacturing have also made it easy to design products, and it is easy to machine such products, where competitors will need specialized tools which are expensive to obtain.

Steel cylinder liner is used to control the bore and surface finishes, which saves on rework wastage and facilitates component integration such as bearings and screws. Whereas the internal combustion (IC) engine is more difficult to produce compared to the electric engine, it offers higher energy density that is favourable to unmanned aerial vehicle (UAV) that needs longer flight time with reduced payload.

Part A Manufacturing

Design Evaluation Table:

PART	CHANGE
CRANKSHAFT	The crankshaft shaft diameter was revised to 4.43 mm, matching the final CAD geometry and bearing selection. This diameter was chosen to satisfy strength requirements while allowing the use of commercially available miniature ball bearings within the limited crankcase space. The crank throw and stroke were retained from the original design. The rear shaft interface was simplified to improve machining accuracy, concentricity, and assembly alignment.
CRANKCASE	The crankcase was locally modified to accommodate the revised 4.43 mm crankshaft diameter and the updated bearing selection. The front and rear bearing seats were resized to ensure proper press fit and alignment. These modifications improve assembly reliability and manufacturability while maintaining the original engine layout.
CONNECTING ROD	The connecting rod length was slightly increased to improve practical assembly clearance and reduce sensitivity to manufacturing tolerances. No changes were made to the joint configuration or overall function of the component.
CYLINDER LINER	The cylinder liner dimensions were refined to match the final crankcase seating depth and ensure correct axial positioning within the engine. The outer diameter and liner length were adjusted to provide a secure press-fit and ensured alignment with the crankcase and cylinder head. The internal bore was retained at 10.49 mm to preserve the validated combustion characteristics.
CARBURETTOR ASSEMBLY	The carburettor layout was revised to improve packaging and accessibility within the compact engine arrangement. The final configuration prioritises simplicity and reliable fuel delivery, with minor geometric adjustments made to ensure proper alignment with the intake port and sufficient clearance from surrounding components.
CONTR-PISTON	The contra-piston geometry was adjusted to suit the final cylinder configuration. The thread profile was refined to improve manufacturability and ease of adjustment. Sealing performance was maintained. Material: 2618 aluminium alloy.
GUDGEION PIN	The gudgeon pin dimensions were revised to suit the updated piston and connecting rod geometry. This improved load distribution and reduced localised stress during oscillating motion. The design supports smooth articulation under operation. The gudgeon pin is manufactured from AISI 52100 bearing steel.
CYLINDER HEAD	Minor geometric refinements improved alignment with the crankcase while maintaining a compact layout. Cooling and structural features were retained with simplified geometry. Material retained as AISI 4340 steel.
BEARINGS	Bearing specification changed from that in the validation report due to the 4.43 mm crankshaft diameter. SKF 619/5-2Z (5×13×4 mm) selected for availability, ease of assembly, and adequate load capacity.

Component Manufacturing

Piston Head:
Manufactured from AISI 4340 steel for high strength and fatigue resistance under combustion loads. The piston head is a very crucial component of the reciprocating part that experiences severe thermal and mechanical forces. It was simplified in geometry and axisymmetric to allow it easy machining, as well as be strong. The materials selection is aimed at wear and thermal resistance in combustion of diesel. The design helps in reducing the amount of time spent machining the design and increasing the level of uniformity when large quantities of production are involved.

Crankshaft:
Manufactured from AISI 4340 alloy steel due to its excellent fatigue strength under torsional and bending loads. There are torsional and bending loads on the crankshaft. The steel AISI 4340 that was selected is owed to its high fatigue strength and resistance. Crankshaft geometry was designed in such a way that one can have it machined by standard procedures and remains structurally good. To reduce the stress concentration in the design and increase the service life, fillets were employed.

Crankcase:
Manufactured from AISI 4340 steel to ensure high stiffness and dimensional stability during operation. It is helped by the crankcase to provide mechanical strength and positioning to the rotating casing. The use of steel AISI 4340 was due to resistance to press-fits and die stress-loads. To make the internal qualities easy, the design has been simplified in order to make it easier to machine and the cost of manufacturing has been minimised. An elevated degree of inflexibility of the steel casing also boosts the well-being of the dimensions throughout the functioning.

Connecting Rod:
Manufactured from AISI 4340 steel to provide strength and fatigue resistance in reciprocating motion. One of the components that transmit forces between the crankshaft and the piston is the connecting rod and thus must be lightweight but powerful. A material made of a steel was utilized in ensuring affordability of fatigue resistance. The geometry of the rod to be used was designed with uniform cross-sections and flowing transitions to improve the manufacturability and reduce the possibilities of failure due to stress.

Cylinder Liner:
Manufactured from hardened steel to provide wear resistance and maintain bore accuracy. Wear-resistant hard bore surface was offered with the help of separately-placed steel cylinder liner. This technique allows proper control of bore geometry and finished surface along with reduced wastage in the event of a possible rework. There is also the benefit of the liner to be removed and the crank case to be removed in order to increase serviceability in the long-term and consistency in production.

Carburettor Assembly:
Manufactured from aluminium alloy for reduced weight. This design presumes that the carburettor connection is an ineffective external simplified component. It is geared towards real-life engine architecture and mounted interfaces modeling. The internal fuel metering units were dropped with aims of reducing complexity in manufacturing and maintaining the exterior geometry in the right position.

Contra-Piston:
Manufactured from AISI 4340 steel to ensure durability and wear resistance during compression adjustment. The contra-piston is used to change the compression in the cylinder. It's made in the miniature form thus being compact, making it easy to produce fan pages and alter. The selection of the materials is with reference to the strength and wear resistance owing to continuous contact of the work.

Gudgeon Pin:
Manufactured from hardened bearing steel for high surface hardness. The gudgeon pin serves to connect the combining rod and the piston and has oscillating loads. It was viewed as high literacy cylindrical fitting and had restricted parameters in order to enhance the straightforwardness of the articulation. The steel material is hardened to minimize the wear and increase the service life of the material.

Cylinder Head:
It closes the combustion chamber with the cylinder head that must withstand pressures and levels of temperatures. This was created in simpler form whose production was not a challenge but with an acceptable sealing quality and strength. The material used was steered towards thermal stability, and wear resistance even in the event of combustion of diesel.

Bearings:
Normal deep groove ball bearings (SKF 619/5-2Z) were selected to aid the crankshaft. This is all evident in the standard parts that reduce cost, purchase is also simplified and the reliability is also improved. This is ensured by the fact that commercially sold bearings were also used in the construction which ensures quality and easy replacement as well.

Step	Component(s) (Material)	Assembly Procedure
1	Crankcase (AISI 4340 steel), Cylinder Liner (Hardened steel)	Check the crankcase–cylinder head unit for cleanliness and dimensional accuracy. Insert the steel cylinder liner into the crankcase bore using an interference fit. Ensure correct axial positioning and concentric alignment to accommodate thermal expansion during operation.
2	Main Bearings (SKF 619/5-2Z, bearing steel)	Seat the bearings into the front and rear bearing housings of the crankcase. Confirm full seating and verify that the bearings rotate freely after installation.
3	Crankshaft (AISI 4340 steel)	Apply light lubrication to the crankshaft journals and bearing inner races. Guide the crankshaft through the installed bearings. Verify smooth rotation and correct axial alignment within the crankcase.
4	Piston Head (AISI 4340 steel), Connecting Rod (AISI 4340 steel), Gudgeon Pin (Hardened bearing steel)	Join the connecting rod to the piston using the gudgeon pin. Check for free articulation at the joint. Insert the piston–rod assembly into the cylinder liner and engage the connecting rod with the crankshaft crankpin.
5	Contra-Piston (AISI 4340 steel)	Thread the contra-piston into the upper cylinder section. Ensure smooth axial movement and correct engagement to allow controlled compression adjustment.
6	Carburettor Assembly (ADC12 aluminium alloy)	Position the carburettor on the intake interface of the crankcase. Align the gravity-fed inlet path and secure the assembly to ensure an airtight and stable connection.
7	Complete Engine Assembly	Inspect all fasteners and interfaces. Verify alignment of rotating components, smooth crankshaft rotation, and correct compression adjustment. Prepare the engine for operational testing at 12,000 rpm.

Sub-Assembly 1:



Figure 1:Piston Explosion.

Sub-Assembly 2:

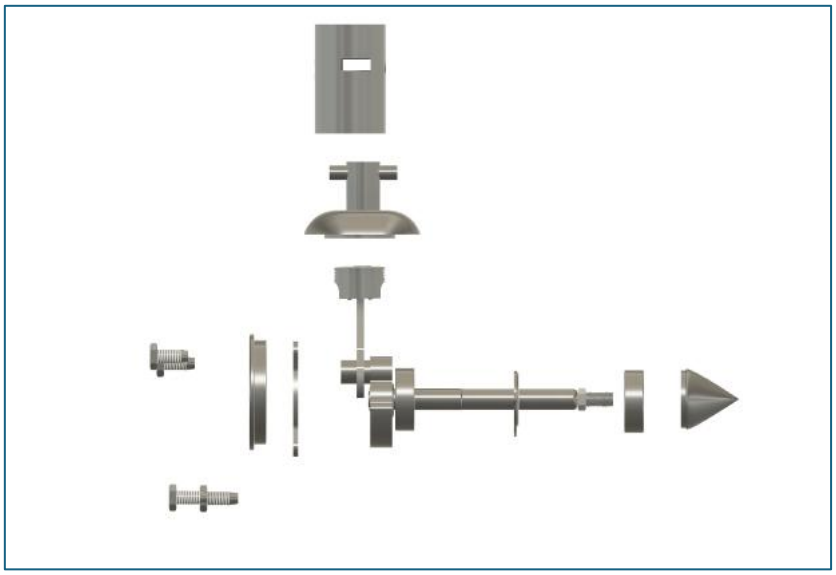


Figure 2: A close up of Primary Components.

Final Assembly:



Figure 3: Individual components in Final Assembly.

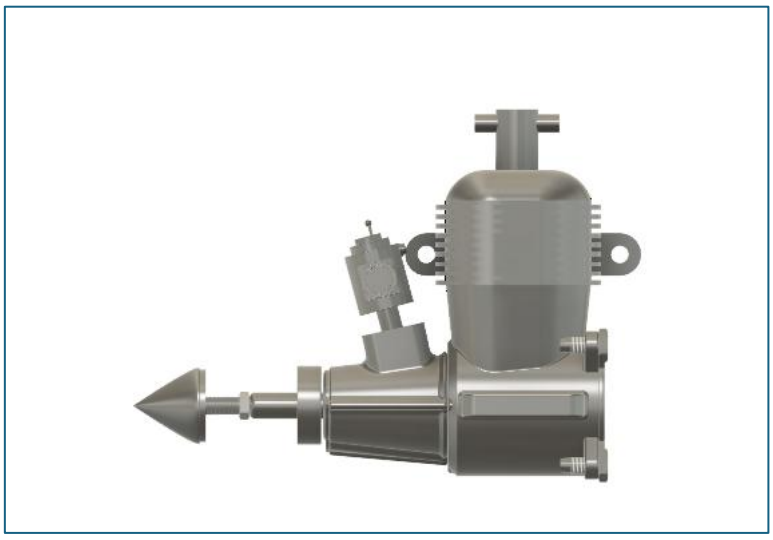


Figure 4: Final Assembly.

Materials Selection

For the purposes of this report, the component chosen for the critical material selection analysis is the piston. The following section defines the key functional requirements that guide the subsequent material evaluation and selection process using CES EduPack.

Critical Analysis Process

Function of a piston:

- Transfers force from combustion gases in the cylinder to the connecting rod
- Converts gas pressure into mechanical motion
- Experiences very high forces over short time periods
- Operates under harsh conditions, which define the main design constraints

For the piston of a miniature two-stroke engine used in a UAV, the functional requirements are:

- High temperature resistance - The piston must withstand combustion temperatures without deformation or softening.
- High mechanical strength and fatigue resistance - It must survive repeated high pressure cycles during operation.
- Low mass - A lighter piston reduces inertia, improves engine response, and increases efficiency.
- Good wear resistance - To maintain performance and avoid seizure against the cylinder walls.
- Adequate thermal conductivity - To transfer heat away quickly and avoid hot spots.
- Controlled thermal expansion - Low expansion ensures consistent clearances with the cylinder and minimal friction.
- Manufacturability & cost efficiency - The material must be produced with reasonable cost and available manufacturing processes.

Initial Screening

The following screening process was conducted using Granta CES EduPack (Level 3 – Advanced) to eliminate unsuitable materials before detailed evaluation.

Stage 1:

A set of initial constraints was defined to eliminate materials that clearly do not meet the essential functional requirements of the piston:

- **Density**: $\leq 5{,}000$ kg/m³
  To ensure a lightweight design and reduce inertial effects.
- **Yield Strength**: ≥ 200 MPa to ensure the material can withstand the mechanical stresses experienced during operation.
- **Maximum Service Temperature**: ≥ 250 °C to ensure the material can tolerate the high temperatures associated with combustion.
- **Thermal Conductivity**: ≥ 50 W/m·K to promote effective heat transfer and prevent hot spots within the piston.

With these constraints, the following graph was created.

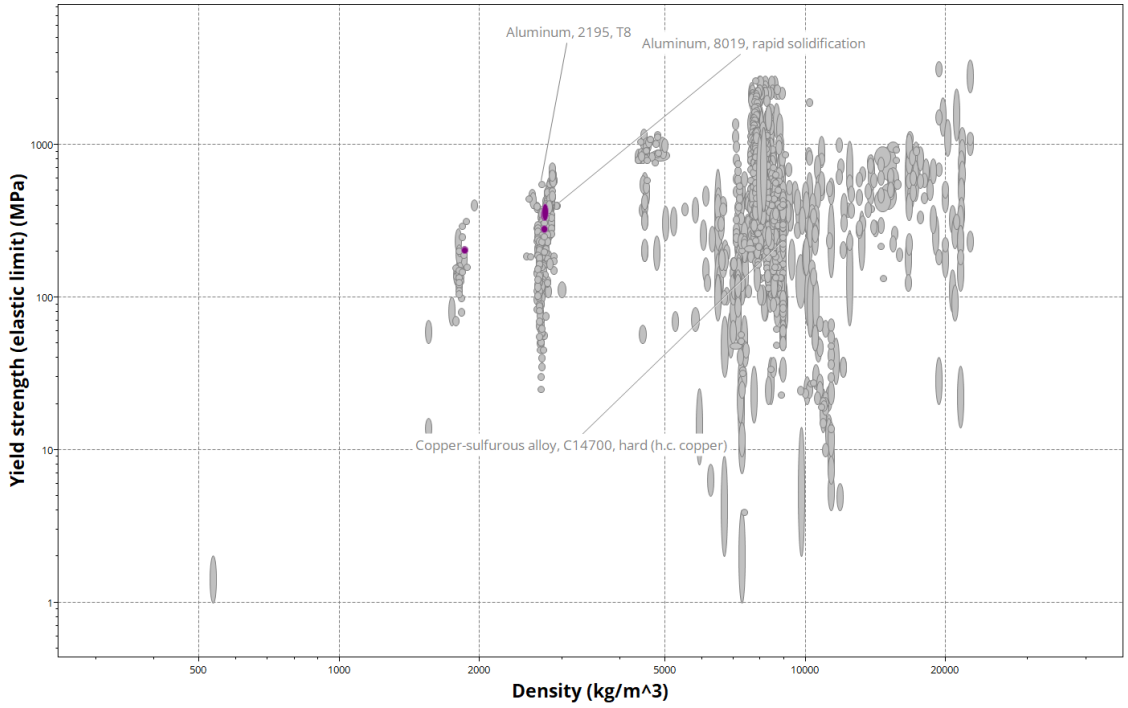


Figure 2.1: Density vs yield strength for materials retained after initial screening

From an initial sample of 1808 metals and alloys in the database, only 12 pass the constraints set above. These can be seen as the coloured circles on the figure and mostly consist of various magnesium and aluminium alloys. These materials stand out for being lightweight while also being strong.

Stage 2:

To further refine the materials retained after the initial screening stage, additional constraints were applied:

- **Hardness (Vickers)**: ≥ 50 HV
  To ensure adequate wear resistance at the piston–cylinder interface during operation.
- **Coefficient of Thermal Expansion**: ≤ 25 µm/m·°C
  To minimise thermal expansion and reduce the risk of piston seizure within the cylinder.

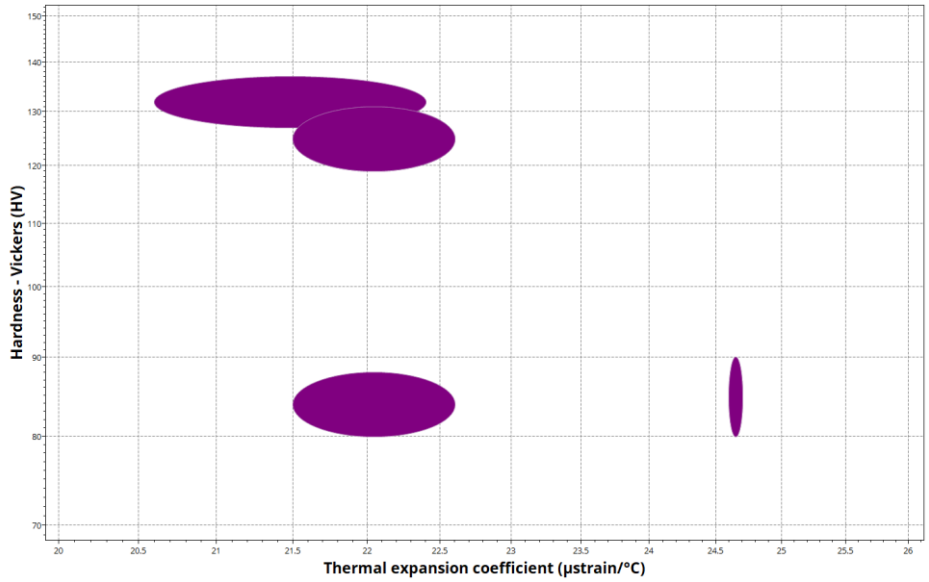


Figure 2.2: Hardness (Vickers) vs thermal expansion coefficient of screened materials

Stage 3:

To ensure economic feasibility for large-scale production, an additional cost-related constraint was applied:

- **Material Cost**: ≤ 10 USD/kg
  To ensure the selected material remains economically viable for a production volume of approximately 50,000 units.

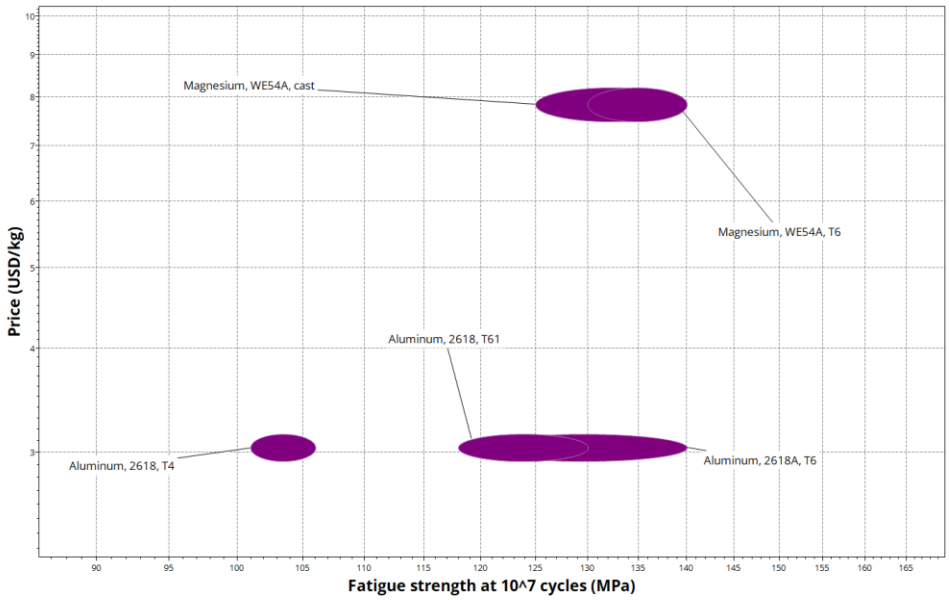


Figure 2.3: Cost versus fatigue strength of screened materials

It is quite clear that Aluminium is the best option by far as it is cheaper than magnesium and most of the options we have are aluminium alloys.

The final material selected for the piston is Aluminium, 2618, T6. Although this exact variant is not explicitly labelled in the screening plots, it is included within the CES EduPack database and meets all

applied screening constraints. This is because it stands out on all performance metrics with its hardness of 137HV, maximum service temperature of 300°C and thermal conductivity of 151W/m·°C. Also, the price per kilogram is USD 3.15. These are impressive numbers which make this material stand out.

Life cycle analysis

Eco Audit:

Using the Eco-Audit tool in Granta EduPack, the analysis of the different phases of life is carried out in a semi-automated manner.
The following input parameters were fed into the system:

- **Product Name:** Piston - Aluminium
- **Quantity:** 50,000
- **Component Name:** Piston
- **Material:** Aluminium, 2618-T6
- **Recycled Content:** 0%
- **Mass:** 0.0066 kg
- **Primary Process:** Metal Rolling & Forging
- **End of Life:** Recycle
- **Transport Type:** Truck (3.5–7.5 t, EURO 3)
- **Distance from factory to warehouse:** 250 km
- **Distance from warehouse to Sharjah Industrial Area 1:** 400 km
- **Product Life:** 4 years
- **Country of Use:** Worldwide

Use:

- **Mobile Mode:** The piston is part of a UAV.
- **Fuel and Mobility Type:** Diesel - Light Goods Vehicle
- **Estimated Usage:** 150 days per year
- **Distance Travelled:** 8 km per day

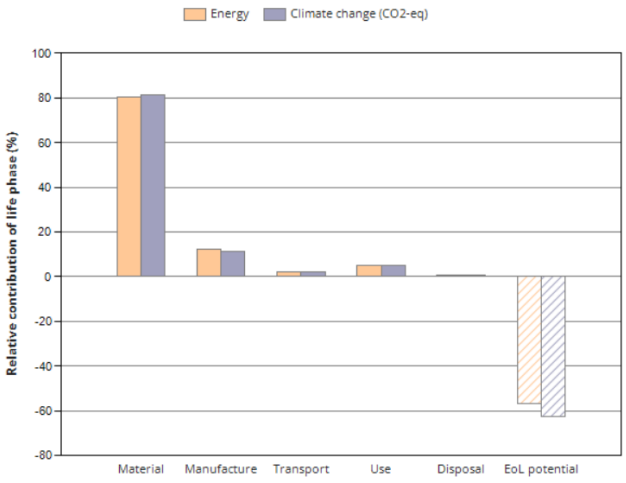


Figure 2.4: Eco-audit results showing the relative contribution of each life-cycle phase

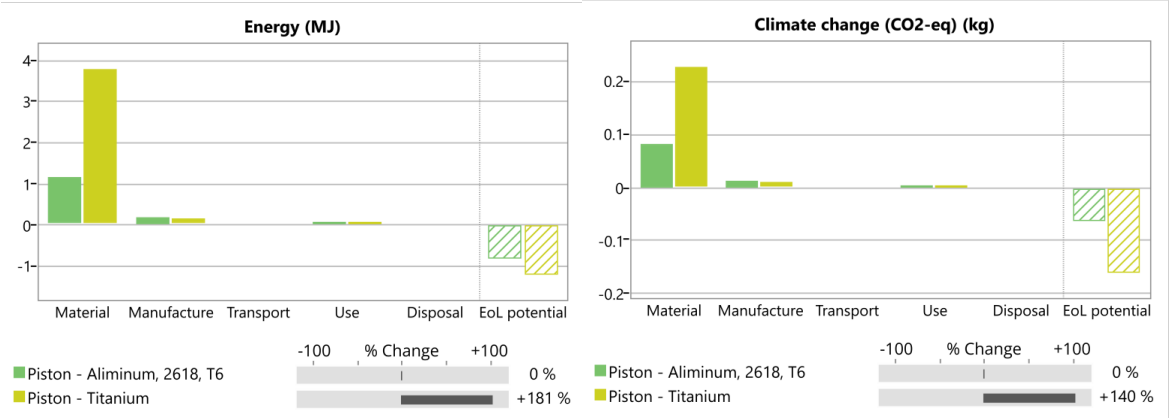
Phase	Energy (MJ)	Energy (%)	Climate change (CO2-eq) (kg)	Climate change (CO2-eq) (%)
Material	1.16	80.3	0.0831	81.3
Manufacture	0.178	12.4	0.0117	11.4
Transport	0.0309	2.1	0.00205	2.0
Use	0.0697	4.8	0.00502	4.9
Disposal	0.00462	0.3	0.000323	0.3
Total (for first life)	**1.44**	**100**	**0.102**	**100**
End of life potential	-0.821		-0.064	

Table 2.5: Energy consumption and carbon footprint by life-cycle phase for the Aluminium 2618 piston

As shown in Table 3.1, the **material production phase is the dominant contributor** to the environmental impact of the Aluminium 2618 piston, accounting for approximately **80% of the total energy consumption** and **over 80% of the total climate change impact (CO_2-eq).** All other life-cycle phases contribute relatively lower in comparison. Despite this, aluminium has a high recycling potential at the end of its life, which helps reduce the total energy use and carbon emissions. This effect is discussed further in the what-if analysis.

What if analysis

The figures below compare the energy consumption and carbon footprint of the selected Aluminium, 2618, T6 piston with a piston made from a certain Titanium alloy.



Figures 2.6 and 2.7: Comparison of energy consumption and carbon emissions for Aluminium 2618 and Titanium pistons

Conclusion

Based on the results of the what-if analysis, a piston made from titanium requires significantly more energy to produce and results in higher carbon emissions compared to the Aluminium, 2618, T6 piston. This difference is mainly due to the higher energy demand associated with titanium material production. In contrast, Aluminium, 2618, T6 shows much lower energy use and carbon footprint across its life cycle. In addition, aluminium benefits from a high end-of-life recycling potential, which helps reduce the overall environmental impact. From an environmental perspective, Aluminium, 2618, T6 is therefore a more suitable and sustainable material choice for the piston in this project.

IC engine vs similar engine

The designed two-stroke combustion engine produces approximately 0.5 kW (0.67 hp) at 12,000 rpm and was compared with a BLDC electric motor of similar power and speed, representative of compact UAV propulsion systems

Component	Combustion Engine	Equivalent Engine
Piston Head	Material: AISI 4340 Steel Manufacturing: CNC machining Details: Intended to be resistant to high thermal and mechanical stresses caused by the diesel combustion process and tolerate its strength and fatigue resistance.	Material: Steel alloy Manufacturing: Machining Details: Uncomplicated piston geometry to equivalent conditions of loading during combustion.
Crankshaft	Material: AISI 4340 Alloy Steel Manufacturing: Forging, CNC machining Details: Able to deal with combined loads in torsion and bending and with better fatigue life.	Material: Steel alloy Manufacturing: Forging, machining Details: Models torque transmission under similar operating conditions.
Crankcase	Material: AISI 4340 Steel Manufacturing: CNC machining Details: Gives it high stiffness and ensures it has accuracy in dimension as it runs in the engine.	Material: Steel housing Manufacturing: Machining Details: Rotating components Structural casing to be used to support rotating components and to align assemblies.
Connecting Rod	Material: AISI 4340 Steel Manufacturing: Forging, machining Details: Vibratory transferring forces, which are of high fatigue strength and durable.	Material: Steel alloy Manufacturing: Forging, machining Details: Is an equivalent transmission of force between crankshaft and piston..
Cylinder Liner	Material: Hardened Steel Manufacturing: Machining, honing Details: Offers wear resistance, smooth bore surface and ensures accuracy of the cylinder.	Material: Hardened steel Manufacturing: Machining Details: Streamlined bore conditions by liner modification to ensure wear resistance.
Gudgeon Pin	Material: Hardened Bearing Steel Manufacturing: Grinding, heat treatment Details: Constructed to take oscillating loads which are high, hard surfaced and wear resistant.	Material: Hardened steel Manufacturing: Grinding Details: Stylized pin denoting the pin rods
Cylinder Head	Material: Steel-based Alloy Manufacturing: Machining Details: Seals combustion chamber and high pressure and high temperature.	Material: Steel alloy Manufacturing: Machining Details: Represents similar sealing and carrying out.
Bearings	Material: Bearing Steel (SKF 619/5-2Z) Manufacturing: Standard component Details: Assures a well-cranking crankshaft with a good load-bearing.	Material: Bearing steel Manufacturing: Standard bearing Details: Gives equal rotational assistance..

Table 3.1 A Comparison of an Internal Combustion Engine Against an Equivalent Electric Motor